

# Homogeneity as presuppositional exhaustification\*

Janek Guerrini and Jad Wehbe<sup>†</sup>

**Abstract** The goal of this paper is to reconcile two observations regarding homogeneity and non-maximality. First, homogeneity is sensitive to constraints on presupposition accommodation, both in the positive case and under negation (Wehbe, 2022). Second, there are asymmetries between positive and negative sentences with definite plurals (Bar-Lev, 2021). We argue that taken together, these two observations support an account of homogeneity in terms of presuppositional exhaustification.

## 1 Introduction

Definite plurals give rise to a truth-value gap known as *homogeneity*. Consider the example in (1): neither (1-a) nor (1-b) are true in a scenario where Mary read only some of the books. The source and pragmatic status of this truth-value gap has received a lot of attention. Early accounts of homogeneity argued that it is a presupposition (Löbner, 2000; Schwarzschild, 1993 a.o.), but several recent accounts have challenged this initial assumption (Križ, 2015; Križ and Spector, 2021; Bar-Lev, 2021).

- (1) a. [Mary read the books] is **true iff** Mary read **all** the books.
- b. [Mary didn't read the books] is **true iff** Mary read **none** of the books.
- c. **Neither is true iff** Mary read only some of the books.

Homogeneity has been argued to come hand-in-hand with non-maximality or exception tolerance, first discussed by Brisson (1998). Consider the example in (2), adapted from Bar-Lev (2021). Here, given a context where what's relevant is whether enough kids were laughing to consider the show a success, (2-b) is judged to be true even if a small portion of the kids weren't laughing. Križ (2015) argues that homogeneity and non-maximality are two sides of the same coin, and indeed other phenomena which give rise to homogeneity effects, such as habituais (Agha, 2021) and conditionals (von Stechow, 1994) also exhibit non-maximality. We therefore take it as a desideratum of a successful account of homogeneity to also account for non-maximality.

- (2) Context: B knew that A was planning a comedy show for kids at a birthday party the day before and was worried about whether it's going to go well.
- a. B asks A: Was the show a success?

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<sup>†</sup>Equal contribution, authors listed in alphabetical order.

- b. A responds: The kids were laughing.

In this paper, we argue for an account of homogeneity and non-maximality in terms of presuppositional exhaustification. Our account builds on Bar-Lev’s implicature account but assumes that implicatures are due to the application of a presuppositional exhaustification operator, *pex* (Bassi et al., 2021; Del Pinal et al., 2024) (cf. Paillé, 2022).

We show that no existing account of homogeneity and non-maximality is able to reconcile the following two observations: (1) homogeneity, both in positive and in negative sentences, is sensitive to constraints on presupposition accommodation (Wehbe, 2022) and (2) empirical evidence suggests (although not completely univocally) that there are asymmetries between positive and negative sentences with definite plurals, for example non-maximal readings are harder to access under negation (cf. Bar-Lev, 2021, but see the discussion below).

We suggest that an account based on *pex* naturally predicts both (1) and (2). At this point, let us observe that on the one hand, the form of our argument is similar to what Del Pinal et al. (2024) argue for Free Choice: the fact that homogeneity behaves like a presupposition receives a natural account within an implicature account assuming *pex*. On the other hand, the arguments we use to argue for the presuppositionality of homogeneity potentially undermine the presuppositionality of Free Choice hypothesized by Del Pinal et al. (2024). In particular, Free Choice does not seem to be sensitive to the constraints on presupposition accommodation that homogeneity and standard presuppositions are sensitive to. Given the parallels that are drawn between homogeneity and free choice in the literature (e.g. Bar-Lev, 2021), this asymmetry raises interesting questions, which we will return to at the end, in section 6.2.

Note that Paillé (2022) argues for a *pex*-based account of homogeneity in definite plurals based on different considerations, but he predicts symmetry in non-maximality (cf. also Paillé, 2023), thus not reconciling (1) and (2). Here, we provide a novel account of this asymmetry in terms of presupposition accommodation, thus accounting for the fact that non-maximality is available but more difficult in negative sentences beyond *pex*, without resorting to any additional mechanisms.

## 2 Bar-Lev (2021) and the asymmetries between positive and negative sentences

Bar-Lev (2021) proposes that definite plurals have an existential meaning. In negative sentences, this existential gets negated and the correct meaning is derived semantically. In positive cases, the existential gets strengthened via an implicature: in a procedure akin to Bar-Lev and Fox’s (2017, 2020) account of Free Choice inferences, the existential gets strengthened to a universal via a process known as Innocent Inclusion of sub-domain alternatives, which are those alternatives that are obtained by replacing the relevant domain variable with its subsets (see, e.g., Chierchia, 2013). Innocent Inclusion was first proposed by Bar-Lev & Fox (2020) as a way to account for Free Choice disjunction, and applies on top of Innocent Exclusion.

- (3) a. **Innocent Exclusion procedure:** (Bar-Lev & Fox, 2020)  
(i) Take all maximal sets of alternatives that can be assigned false consistently with the prejacent.

- (ii) Only exclude (i.e., assign false to) those alternatives that are members in all such sets—the Innocently Excludable (=IE) alternatives.
- b. **Innocent Inclusion procedure:** (Bar-Lev & Fox, 2020)
  - (i) Take all maximal sets of alternatives that can be assigned true consistently with the prejacent and the falsity of all IE alternatives.
  - (ii) Only include (i.e., assign true to) those alternatives that are members in all such sets — the Innocently Includable (=II) alternatives.
- (4) Application to Free Choice Disjunction: (Bar-Lev & Fox, 2020)
  - a.  $\diamond(a \vee b)$
  - b. Alternatives:  $\{\diamond a, \diamond b, \diamond(a \wedge b), \diamond(a \vee b)\}$
  - c. Innocently excludable alternatives: all those alternatives that can be negated without affecting the truth of the prejacent, i.e.  $\diamond(a \wedge b)$
  - d. In this case, including all the alternatives which are not IE is consistent with the truth of  $\diamond(a \vee b)$  (the prejacent) and the falsity of  $\diamond(a \wedge b)$  (the IE alternative).
  - e. This means we only have one maximal set of alternatives to consider,  $\diamond(a \vee b), \diamond a, \diamond b$ , and consequently all alternatives within this set are II.
  - f. Thus,  $exh^{IE+II}[\diamond(a \vee b)] = \diamond(a \vee b) \wedge \diamond a \wedge \diamond b \wedge \neg \diamond(a \wedge b)$

Bar-Lev (2020) proposes an account of homogeneity that relies on a parallel mechanism: we will present his idea more in detail below, but in a nutshell, sentences containing a definite plural are underlyingly existential, as in (5-a), and after innocent inclusion of sub-domain alternatives the existential is strengthened to a universal (5-c):

- (5) a.  $\llbracket \text{the kids smiled} \rrbracket^w = \exists x \leq \llbracket \text{the kids} \rrbracket^w : x \in D \wedge \llbracket \text{smiled} \rrbracket^w(x)$
- b. Alternatives:
 
$$\{\lambda w'. \exists y \leq_{AT} \llbracket \text{the students} \rrbracket^w : y \in D' \wedge \llbracket \text{smiled} \rrbracket^{w'}(y) \mid D' \subseteq D \wedge D' \cap \llbracket \text{the students} \rrbracket^w \neq \emptyset\}$$
- c.  $\forall A. A \in \text{ALT} \wedge A \in \text{II} \rightarrow A(w) = 1$ , i.e.
 
$$\llbracket exh \text{ the kids smiled} \rrbracket^w = \forall x \leq \llbracket \text{the kids} \rrbracket^w : x \in D \rightarrow \llbracket \text{smiled} \rrbracket^w(x)$$

In appropriate cases, exhaustification can ignore contextually irrelevant alternatives, a process known as pruning: in Bar-Lev’s account, some of the sub-domain alternatives can be pruned, which results in non-maximal readings. In other words: some sub-domain alternatives are not considered, and thus do not get included, and therefore the end truth conditions are weaker-than-universal.

This mechanism is not available to derive non-maximality in negative sentences, as there the sub-domain alternatives are weaker than the prejacent and therefore exhaustification does not strengthen the meaning. For these cases, Bar-Lev proposes a different, more costly mechanism. This makes for the feature that has sparked interest for Bar-Lev’s account (see, e.g., Chierchia, 2022): it predicts positive and negative sentences to behave asymmetrically.

Bar-Lev argues that this prediction captures at least two empirical patterns. Firstly, controlling for context, non-maximal readings are more easily available in positive cases than under negation. For example, in the context in (6), (6-a) allows for a non-maximal reading where it

suffices that every kid took at least 2 vitamins. On the other hand, with the negative quantifier in (6-b) the analogous non-maximal reading (*no kid took at least 2 vitamins*) is not available (examples from Bar-Lev, 2021).

- (6) Context: the kids are required to take at least 2 of the 4 vitamins I gave each of them.
- a. All of the kids took their vitamins. = *Every kid took at least 2 vitamins.*
  - b. None of the kids took their vitamins. = *No kid took any vitamin (i.e., not even one vitamin).*

This is partly confirmed experimentally by recent work by Augurzky *et al.*, who compared positive sentences like *Every boy opened his presents* to negative sentences like *No boy opened his presents* across different types of contexts. In contexts not favouring non-maximality, exceptions were taken by participants to make false both positive and negative sentences. However, in contexts favouring non-maximality, positive sentences, but not negative sentences were judged true in the presence of exceptions. We therefore take this observed asymmetry in non-maximality between upward-entailing and downward-entailing environments to be a desideratum of a successful account of homogeneity and non-maximality.<sup>1</sup>

Secondly, experimental evidence by Tieu *et al.* (2019), looking at 4 and 5 year-old French speaking children, shows that while some children accept sentences like ‘the hearts are red’ as true in a situation where some but not all of the hearts are red (Karmiloff-Smith 1981; Caponigro *et al.* 2012), no children accept sentences like ‘the hearts aren’t red’ as true in such a situation. This, Bar-Lev argues, is explained if we assume that children are not computing the implicature in the positive case.

A last experimental result worth mentioning in this direction is Križ and Chemla (2015), who reveal a marked asymmetry for homogeneity-triggering sentences. In conditions where not all objects had the target feature (e.g., not all triangles were green), most participants judged “The triangles are green” as not completely false or not completely true, whereas “The triangles aren’t green” was deemed completely false in about half of the responses – over twice as many as in the positive case. We discuss this result in more detail in section 5.1.

### 3 The non-at-issueness of homogeneity

Wehbe (2022) uses a constraint on presupposition accommodation to argue that homogeneity is in fact a presupposition. In particular, homogeneity has been argued to differ from standard presuppositions due to its apparently distinct projection patterns (Spector, 2013; Križ, 2015). For example, while presuppositions project from the antecedents of conditionals (7-b), homogeneity does not project from this same environment (7-a).

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<sup>1</sup>Note that Augurzky *et al.* tested another downward-entailing environment, *not every*, in addition to *no*, and surprisingly, *not every* patterns with *every* rather than *no*. This is a problem for implicature-based theories such as ours or Bar-Lev’s. We leave this as an open problem. Notice, on the other hand, that for theories predicting symmetry (such as Križ’s), non-maximality in the “not every” context is entirely expected – but conversely, on those theories the relative lack of non-maximality in the “no” context is unexpected.

- (7) a. If Mary read the books, she passed the exam.  
       ↗ Mary read all or none of the books.  
       b. If it is Mary who read the books, she passed the exam.  
           → Someone read the books.

One confound in these diagnostics is the availability of local accommodation. For example, given (7-a), it is possible that homogeneity is in fact a presupposition but that it is locally accommodated in the antecedent of the conditional, thus explaining the lack of projection. Of course one still has to explain the difference between (7-a) and (7-b), but this confound calls for a diagnostic that does not rely on projection. Doron and Wehbe (2022) argue for such a diagnostic based on a constraint on accommodation first proposed by Orin Percus and Irene Heim in lecture notes.

Doron and Wehbe argue that conditions on assertability offer a way to detect the division of labor between presupposition and assertion. In particular, assertions must not be trivial (Stalnaker, 1974), and given that anti-triviality is a constraint on assertions, it is evaluated with respect to the common ground after presupposition accommodation. We therefore end up with the principle in (8). Empirical evidence for this principle comes from contrasts such as in (9). After collapsing presupposition and assertion, (9-a) and (9-b) convey the same information, yet (9-a) is infelicitous. The second sentence in (9) presupposes that John has 10 children, but accommodating this presupposition gives rise to infelicity. This is because after accommodating the presupposition, the assertion becomes trivial, violating PAI.

- (8) **Post-Accommodation Informativity (PAI):** A sentence  $S_p$  (presupposing  $p$ ) can be uttered felicitously only if  $S_p$  is informative w.r.t the common ground after accommodating  $p$ .
- (9) a. I knew that all of John's kids are adopted but today I discovered something amazing.  
       # All 10 of John's kids are adopted!  
       b. I knew that all of John's kids are adopted but today I discovered something amazing.  
           John has 10 adopted kids!

We can therefore use PAI to test whether homogeneity is a presupposition without relying on projection. Consider the examples in (10). If homogeneity was part of the asserted meaning, as in Bar-Lev's account, there is no reason to expect any infelicity in (10-a) and (10-b). On the other hand, this infelicity is predicted by PAI assuming that homogeneity is a presupposition. In (10-a), given that it is common ground that Mary read at least some of the books, once we accommodate the homogeneity presupposition in (11), the second sentence becomes trivial. Similarly in (10-b), the second sentence is trivial once we accommodate homogeneity, given that it is known that Mary didn't read all of the books.

- (10) a. I knew that Mary read at least some of the books, but today I learned something interesting. # She read the books.  
       b. I knew that Mary didn't read all of the books, but today I learned something interesting. # She didn't read the books.
- (11) **Homogeneity presupposition:** Mary read either none or all of the books.

One might consider an alternative explanation of (10-a-b) in terms of focus, where the dis-

course structure requires that stress should be placed on an element in contrast to the ‘some’ appearing in the initial sentence (for an implementation of an alternative along these lines, see Bassi & Bar-Lev, 2024). To wit, on such a view since the second sentences in (11) lack an overt element that can bear stress and contrast with *some*, they do not meet this requirement.<sup>2</sup> However, we would like to note that the focus explanation not extend straightforwardly to parallel cases where the information like that contained in the first sentences of (10) and (11) is not asserted explicitly in the same discourse as our target examples. In (12-b) (Bassi, p.c., attributed to Roger Schwarzschild), it is in fact common knowledge that John isn’t married to all of his teammates, thus similarly making the sentence trivial after accommodating homogeneity. Similarly, in (13), it is common ground that at most one candidate can win, thus making (13-c,d) both trivial after accommodating homogeneity. Notice that the counterparts with quantifiers in (12-a) and (13-a,b) convey the same information and are felicitous in the relevant contexts.

- (12) a. John isn’t married to any of his teammates.  
b. #John isn’t married to his teammates.
- (13) Context: The rules for this year’s election are that the person with the most votes wins, but if there is a tie for most votes, all of the candidates lose. This year..  
a. None of the candidates won.  
b. All of the candidates lost.  
c. #The candidates didn’t win.  
d. #The candidates lost.

Bassi and Bar-Lev (2024) argue that examples like (12-b) and (13-c) can be accounted for on Bar-Lev’s (2021) assertive implicature account of homogeneity, but their explanation does not extend to the positive counterparts like (13-d). In short, they assume, following Magri (2011), that implicatures are obligatorily derived when they result in a contradiction, even when the implicature is triggered locally in a downward entailing environment. In (12-b), the implicature results in strengthening from *John is married to some of his teammates* to *John is married to all of his teammates*, which contradicts the common ground. Therefore this implicature is obligatorily computed under negation and the resulting parse after negating the contradiction is a contextual tautology and is therefore predicted to be infelicitous. While we will not extensively compare this alternative account of (12-b) to the one on terms of presupposition accommodation, we would like to point out that the account in terms of a Magri effect does not extend to the positive case in (13-d). Here, an assertive *exh* account of homogeneity would predict that (13-d) simply asserts that all of the candidates lost, and it is therefore not clear why there should be a difference between (13-d) and (13-b) on such an account. Of course, one can stipulate that *exh*(p) is infelicitous when p is entailed by the context, but the presuppositional account offers a systematic explanation for this in terms of independently needed constraints on presupposition accommodation. We therefore take the data in this section to show that homogeneity is sensitive to constraints on presupposition accommodation.<sup>3</sup>

<sup>2</sup>We also thank the editor for pushing us to think about this alternative.

<sup>3</sup>Note that while we focus on a particular constraint on presupposition accommodation to argue that homogeneity is a presupposition, there are presumably other pragmatic differences between presuppositions and assertions that might result in different conditions on when sentences with homogeneity and counterparts with *all* cancelling homogeneity can be used. For example, consider (i), which is adapted from examples provided to us by

## 4 Applying *pex* to homogeneity

At this point, let us observe that accounts viewing homogeneity as arising from a lexical presupposition (Schwarzschild, 1993 a.o.) account well for the non-at-issueness of homogeneity laid out in section 3. However, since they are symmetric, these accounts are unable to account for the asymmetries between positive and negative sentences. Conversely, Bar-Lev captures this asymmetry, but the data in (10) poses a problem for his account. Given that exhaustification simply strengthens the asserted meaning, it is not clear why the second sentence in (10-a) is infelicitous, since its strengthening provides new information (i.e., that Mary read *all* of the books). Even granting that there is some mechanism by which we get infelicity when the proposition is trivial before applying *Exh*, it is not clear why (10-b) is infelicitous, since in this case it is the literal meaning that provides new information (i.e., that Mary didn't read *any* book).

In what follows, we lay out our account, which combines ingredients from presuppositional theories and from Bar-Lev's implicature theory. Following Bar-Lev (2020), we propose that there is an existential pluralization operator which is always present when a distributive predicate is predicated of a plural individual. The lexical entry, slightly modified from Bar-Lev (2020), is given in (14): the operator takes a domain variable which must include all subparts of the definite plural, a 1-place predicate and an individual and returns true if the predicate is true of some atomic part of the individual.

$$(14) \quad \llbracket \exists\text{-PL} \rrbracket^w = \lambda D_{\langle e, t \rangle}. \lambda P_{\langle e, t \rangle}. \lambda x. \exists y \leq_{AT} x : y \in D \wedge P(y) = 1$$

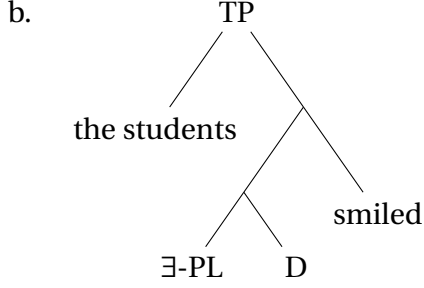
Consider an example derivation in (15). The basic truth conditions for *the students smiled* return true if at least one of the students smiled.

- (15) a. The students smiled.

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two anonymous reviewers. Here, (i-a) is infelicitous even though, after accommodating homogeneity, the sentence is nevertheless not trivial (it asserts that John is married to all rather than none of his teammates). Moreover, the fact that (i-b) is acceptable suggests that we can accommodate a common ground in (i) where it is possible that John is married to all of his teammates and such a common ground should make (i-a) also acceptable. Spector (2024) argues based on similar examples with *some*, that implicatures are ruled out when the strengthened meaning with the implicature is much more informative/controversial than the parse without the implicature. This applies to our case here: the basic meaning of (i-a) (*John is married to some of his teammates*) is not controversial since it is compatible with John being married to only one of his teammates. On the other hand, the strengthened meaning (*John is married to all of his teammates*), is very controversial since it contradicts a very likely assumption that John can't be married to more than one of his teammates. While Spector (2024) uses this type of example as an argument for a different account of implicatures not as presuppositions, Doron et al. (2024) argue that this constraint in fact also follows from independently needed constraints on presupposition accommodation, assuming that implicatures are due to *pex*. Since this additional test is tangential to the main point of our paper, we do not go through the account here, but if we accept Doron et al.'s arguments, then (i) can be accounted for within a presuppositional account of homogeneity.

- (i) It's crazy, I didn't realize this was possible. Somehow..  
 a. # John married his teammates.  
 b. John married all of his teammates.



- c.  $\llbracket \text{TP} \rrbracket^w = 1$  iff  $\exists y \leq_{AT} \llbracket \text{the students} \rrbracket^w : y \in D \wedge \llbracket \text{smiled} \rrbracket^w(y)$   
 $= 1$  iff  $\exists y \leq_{AT} \llbracket \text{the students} \rrbracket^w : \llbracket \text{smiled} \rrbracket^w(y)$  (since D contains all atomic parts of  $\llbracket \text{the students} \rrbracket$ ).

We propose that homogeneity arises from the presuppositional exhaustification operator *pex*, first proposed by Bassi et al. (2021) and Del Pinal et al. (2024), and defined below.<sup>4</sup>

(16)  $\llbracket \text{pex}[S] \rrbracket^w :=$

a. **asserts:**

$\llbracket S \rrbracket^w$

b. **presupposes:**

(i)  $\bigwedge \neg p(w)$ : *p* is innocently excludable

(ii)  $\forall q (q \text{ is innocently includable} \wedge q \in R \rightarrow q(w) = 1) \vee$

$\forall q (q \text{ is innocently includable} \wedge q \in R \rightarrow q(w) = 0)$

where *R* = a contextually assigned ‘relevance’ predicate which satisfies the following conditions:

(a) the prejacent,  $\lambda w. \llbracket S \rrbracket^w$ , is relevant (i.e.,  $\lambda w. \llbracket S \rrbracket^w \in R$ ), and

(b) any proposition that is contextually equivalent to the prejacent is also in *R* (i.e., if  $\lambda w. \llbracket S \rrbracket^w \cap c \equiv \psi \cap c$ , then  $\psi \in R$ )

Del Pinal *et al.* (2024)

*Pex* presupposes (i) the negation of all innocently excludable alternatives, as shown in (16-b). For innocently includable alternatives, it (ii) presupposes that they are either all true and all false.

While the definition of *pex* in (16) makes it look like homogeneity is being in a way stipulated in the definition of innocent inclusion, we believe that there is a uniform way of to define *pex* in terms of assertive *exh* that encompasses both innocent exclusion and innocent exclusion (Itai Bassi, p.c). This definition is given in (17) and gives the same results as the one in (16) for the purposes of the cases discussed in this paper. Since this issue is independent of the main point of our paper, we stick with the definition given in Del Pinal et al. (2024) to illustrate our derivations here.

<sup>4</sup>Note that we use a presuppositional exhaustification operator with innocent inclusion in order to derive homogeneity, thus implementing Bar-Lev’s account within a *pex*-based theory. There is a question here about whether our proposal can be recast in terms of recursive exhaustification, with a *pex* operator that only has innocent exclusion. In fact, Chatain (2021) proposes an analysis of homogeneity in terms of recursive application assertive *exh* with only innocent exclusion, and Del Pinal et al. (2024) offer a way to account for free choice in terms of recursive *pex*. As far as we can see, we could therefore potentially account for homogeneity in terms of recursive *pex* in a parallel way to what Del Pinal et al. (2024) suggest for free choice.

- (17)  $\text{pex}_{IE+II}(\llbracket \phi \rrbracket^w) =$   
 a. Presupposition:  $\llbracket \phi \rrbracket^w \rightarrow \text{exh}_{IE+II}(\llbracket \phi \rrbracket^w)$   
 b. Assertion:  $\llbracket \phi \rrbracket^w$

At this point, we can derive the universal meaning for (15-a). We assume that *pex* obligatorily applies at the highest node. In line with Bar-Lev (2021), we assume that the alternatives for (15-a) are the subdomain alternatives, which consist of replacing *D* in (15-b) with its subsets. As discussed by Bar-Lev (2021) and Chierchia (2013), in order to get the required strengthening, we have to assume that alternatives where the intersection with  $\llbracket \text{the students} \rrbracket$  is empty are not considered. We therefore end up with the alternative set in (18) for (15-a).<sup>5</sup>

- (18) Alternatives for *pex* in [*pex* [*the students smiled*]] at the evaluation world *w*.  
 (We will use  $\text{Alt}_{\text{pex}}$  as shorthand):  
 $\{\lambda w'. \exists y \leq_{AT} \llbracket \text{the students} \rrbracket^w : y \in D' \wedge \llbracket \text{smiled} \rrbracket^{w'}(y) \mid D' \subseteq D \wedge D' \cap \llbracket \text{the students} \rrbracket^w \neq \emptyset\}$

Given this set of alternatives applying *pex*, as defined in (16), gives us the result in (19). The subdomain alternatives are not innocently excludable, but they are innocently includable, which gives rise the homogeneity presupposition in (19-b). In short, we predict that *the students smiled* presupposes that either all or none of the student smiled and is true if all of them smiled. This gives us the universal meaning for (15-a), after collapsing presupposition and assertion. This can be seen clearly by applying the A operator to (19), as shown in (20).<sup>6</sup>

- (19)  $\llbracket \text{pex}[\text{The students smiled}] \rrbracket^w$   
 a. **asserts:**  
 $\exists x \leq_{AT} \llbracket \text{the students} \rrbracket^w : x \in D \wedge \llbracket \text{smiled} \rrbracket^w(x)$   
 b. **presupposes** (as far as sub-domain alternatives are concerned, here and henceforth):  
 $[\forall q. q \in R \wedge q \in \{\lambda w'. \exists x \leq_{AT} \llbracket \text{the students} \rrbracket^w : x \in D' \wedge \llbracket \text{smiled} \rrbracket^{w'}(x) \mid D' \subseteq D \wedge D' \cap \llbracket \text{the students} \rrbracket^w \neq \emptyset\} \rightarrow q(w) = 1 \vee$   
 $[\forall q. q \in R \wedge q \in \{\lambda w'. \exists x \leq_{AT} \llbracket \text{the students} \rrbracket^w : x \in D' \wedge \llbracket \text{smiled} \rrbracket^{w'}(x) \mid D' \subseteq D \wedge D' \cap \llbracket \text{the students} \rrbracket^w \neq \emptyset\} \rightarrow q(w) = 0]$   
 $\models$   
 $\forall x : x \leq_{AT} \llbracket \text{the students} \rrbracket^w \rightarrow \llbracket \text{smiled} \rrbracket^w(x) \vee \neg \exists x : x \leq_{AT} \llbracket \text{the students} \rrbracket^w \wedge \llbracket \text{smiled} \rrbracket^w(x)$   
 (modulo pruning)

- (20)  $A(\llbracket \text{pex The students smiled} \rrbracket) = 1$  iff  $\forall x : x \leq_{AT} \llbracket \text{the students} \rrbracket \rightarrow \llbracket \text{smiled} \rrbracket(x)$

<sup>5</sup>A reviewer observes that there is a genuine question of the world at which the noun is evaluated at. For simplicity, we here assume that the denotation of the noun is constant across the context set and that the noun is evaluated at the world of evaluation. As mentioned by the reviewer, this raises an issue regarding the precise derivation of subdomain alternatives, which is typically overlooked in the literature and remains tangential to our focus here.

<sup>6</sup>The A-operator is defined in (i): it takes as an input a truth value and returns 1 if its input is 1, and 0 if its input is 0 or # – i.e. it returns 1 if both its presupposition and assertion are true and 0 if the relevant proposition is undefined or false (i.e. either the presupposition is not true or the presupposition is true but the assertion isn't).

(i)  $\llbracket A \rrbracket(t) = \begin{cases} 1 & \text{iff } t = 1 \\ 0 & \text{iff } t = \# \vee t = 0 \end{cases}$  Bochvar (1981)

This straightforwardly captures the PAI data from Wehbe (2022) in the positive case: given a context where it is common ground that some of the students smiled, accommodating the homogeneity presupposition makes the assertion trivial, thus violating PAI.

Finally, we are able to get non-maximality in the same way as Bar-Lev (2020), by pruning alternatives. Crucially here, pruning makes the homogeneity presupposition weaker which results in non-maximal truth conditions once this presupposition is accommodated. To see this with a toy example, consider the model with only three students in (21) and suppose that we prune all alternatives except for the ones where the domain includes exactly 2 of the students (21-b). The weakened homogeneity presupposition would be the one given in (21-c). After collapsing presupposition and assertion, we predict that given the R in (21-b), *the students smiled* is true if at least 2 out of the three students smiled. We therefore derive non-maximality within the *pex* account in a parallel way to what Bar-Lev does with *exh*.<sup>7</sup>

- (21) a.  $\llbracket \text{the students} \rrbracket = a \oplus b \oplus c$   
 b.  $R = \{\lambda w. \exists x \leq_{AT} y : \llbracket \text{smiled} \rrbracket^w(x) \mid y \in \{a \oplus b, b \oplus c, a \oplus c\}\}$   
 c.  $[\exists x \leq_{AT} a \oplus b : \llbracket \text{smiled} \rrbracket(x) \wedge \exists x \leq_{AT} b \oplus c : \llbracket \text{smiled} \rrbracket(x) \wedge \exists x \leq_{AT} a \oplus c : \llbracket \text{smiled} \rrbracket(x)] \vee$   
 $[\neg \exists x \leq_{AT} a \oplus b : \llbracket \text{smiled} \rrbracket(x) \wedge \neg \exists x \leq_{AT} b \oplus c : \llbracket \text{smiled} \rrbracket(x) \wedge \neg \exists x \leq_{AT} a \oplus c : \llbracket \text{smiled} \rrbracket(x)]$   
 d. **Final TC after accommodation:** 1 iff at least two out of the three students smiled.

Turning to negative sentences, Bar-Lev assumes that no exhaustification applies in this case and the negated existential meaning delivers the correct truth conditions, as shown in (22). In this case, even if *exh* were to apply above negation, its contribution would be trivial since the subdomain alternatives are weaker than the prejacent.

- (22) a. The students didn't smile.

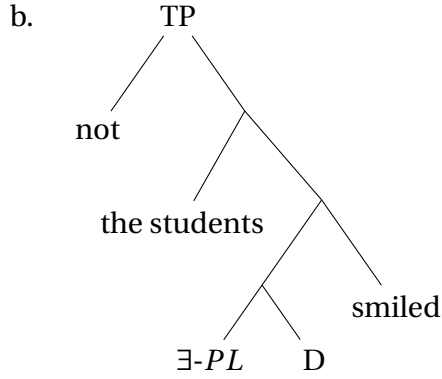
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<sup>7</sup>It should be mentioned that we also assume the same constraints on pruning as Bar-Lev, i.e. the constraint on pruning in (i) due to Crnič *et al.* (2015).

- (i) (cf. Crnič *et al.*, 2015)  
 Constraint on pruning (1st version):  
 $Exh_C(S)$  is licensed for  $C \subseteq Alt(S)$  only if  $\llbracket Exh_{Alt(S)} S \rrbracket \Rightarrow \llbracket Exh_C S \rrbracket$ .

Effectively, this means that pruning can only lead to weakening.

This is needed for the following reason. We have seen that by presuppositionally exhaustifying over a subset of the alternatives, as in (21), we derive a non-maximal interpretation equivalent to “at least two of the students smiled”. Naturally, various other readings can emerge by selecting different subsets from the full set of alternatives. However, some readings that could theoretically be derived by selecting certain subsets of alternatives are not observed. For example, if we prune all alternatives except “John smiled,” this alternative would become IE, leading to the interpretation “(Jane smiled  $\vee$  Bill smiled)  $\wedge$   $\neg$  John smiled.” However, no context supports an interpretation where “the students smiled” implies that one of them didn't smile. While the meaning may be compatible with some students not smiling (non-maximality), there is no evidence that it can entail such a scenario, a fact that is captured by the constraint on pruning in (i).



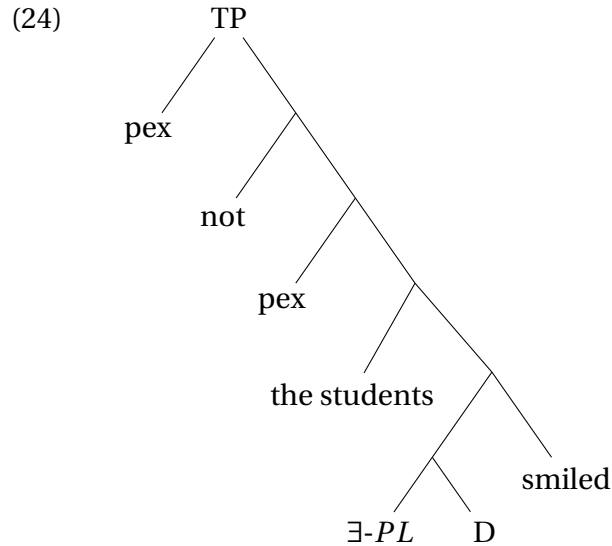
c.  $\llbracket \text{TP} \rrbracket = 1$  iff  $\neg \exists y \leq_{AT} \llbracket \text{the students} \rrbracket : y \in D \wedge \llbracket \text{smiled} \rrbracket(y)$

While the correct truth conditions are predicted if we assume that *pex* doesn't apply here, the data with PAI would remain unexplained. Consider (23): if the LF in (22-b) were the correct one for (22), there would be no presupposition and therefore no reason to expect a PAI effect.

(23) I knew that not all of the students smiled, but today I learned something interesting. #  
The students didn't smile.

We will argue that (23) illustrates a case where, somewhat surprisingly, the contribution of *pex* can be detected even though it doesn't affect the overall truth conditions after collapsing presupposition and assertion. This is a conclusion also argued for by Wehbe and Doron (2024) looking at implicatures with *or* under negation.

We assume, following Magri (2011), that exhaustification applies not only at the highest node, but also, obligatorily, at every scope site below it, as shown in (24). Therefore *pex* has to be inserted below negation, giving rise to the same homogeneity presupposition we derive for the positive case in (19). This presupposition will project from the scope of negation, giving rise to the truth conditions in (25) below the higher *pex*.



(25)  $\llbracket \text{not}[\text{pex}[\text{the students } \exists\text{-PL } D \text{ smiled}]] \rrbracket^w =$

a. **asserts:**

- $\neg\exists x \leq \llbracket \text{the students} \rrbracket^w : \llbracket \text{smiled} \rrbracket^w(x)$
- b. **presupposes:**  
 $\forall x : x \leq_{AT} \llbracket \text{the students} \rrbracket^w \rightarrow \llbracket \text{smiled} \rrbracket^w(x) \vee \neg\exists x : x \leq_{AT} \llbracket \text{the students} \rrbracket^w \wedge \llbracket \text{smiled} \rrbracket^w(x)$   
(modulo pruning)

This gives correct truth conditions for ‘The students didn’t smile’. However, given the assumption that *pex* applies at every scope site, another *pex* has to apply above negation. In order to determine what the contribution of this higher *pex* will be, we have to consider how *pex* deals with cases when its prejacent and alternatives have presuppositions. We assume that the innocent inclusion presupposition, restated in (26), targets truth and falsity such that in both disjuncts the presupposition of  $\alpha$  has to be satisfied. Additionally, the presupposition of the prejacent projects from the scope of *pex*.

$$(26) \quad \begin{aligned} &\forall \alpha (\alpha \text{ is innocently includable} \wedge \alpha \in R \rightarrow \alpha(w) = 1) \vee \\ &\forall \alpha (\alpha \text{ is innocently includable} \wedge \alpha \in R \rightarrow \alpha(w) = 0) \end{aligned}$$

We can now compute our final truth conditions for (24). Romoli (2012) shows that there can be no alternatives in which negation is deleted. Additionally, we assume that the lower *pex* cannot be deleted in the alternatives. The only alternatives for the higher *pex* are therefore the subdomain alternatives given in (27-d). Given these assumptions, the higher *pex* does not affect the truth conditions and innocent inclusion again simply gives a homogeneity presupposition, as shown in (27-c).

- (27) a.  $\llbracket \text{pex}_1[\text{not}[\text{pex}_2[\text{the students } \exists\text{-PL D smiled}]]] \rrbracket^w =$   
b. **Assertion:**  $\neg\exists x \leq \llbracket \text{the students} \rrbracket^w : \llbracket \text{smiled} \rrbracket^w(x)$   
c. **Presupposition:**  $\forall D' \subset D : D' \cap \llbracket \text{the students} \rrbracket^w \neq \emptyset \rightarrow (\neg\exists x \leq_{AT} \llbracket \text{the students} \rrbracket^w : x \in D' \wedge \llbracket \text{smiled} \rrbracket^w(x) \vee \forall x \leq_{AT} \llbracket \text{the students} \rrbracket^w : x \in D' \rightarrow \llbracket \text{smiled} \rrbracket^w(x))$   
 $\Leftrightarrow \forall x : x \leq_{AT} \llbracket \text{the students} \rrbracket^w \rightarrow \llbracket \text{smiled} \rrbracket^w(x) \vee \neg\exists x : x \leq_{AT} \llbracket \text{the students} \rrbracket^w \wedge \llbracket \text{smiled} \rrbracket^w(x)$   
d.  $\text{Alt}_{\text{pex}_1} = \{\lambda w. \llbracket \text{not}[\text{pex}_2[\text{the students } \exists\text{-PL D' smiled}]] \rrbracket^w \mid D' \subseteq D \wedge D' \cap \llbracket \text{the students} \rrbracket^w \neq \emptyset\}$

As shown in (28), the truth conditions after accommodation are identical to those in (22-c). Nevertheless, since there is a homogeneity presupposition now, we predict the PAI effect in (23): after accommodating homogeneity, the second sentence in (23) becomes trivial, violating PAI. We have therefore accounted for the symmetric PAI effects while maintaining an asymmetric account of homogeneity in terms of *pex*.

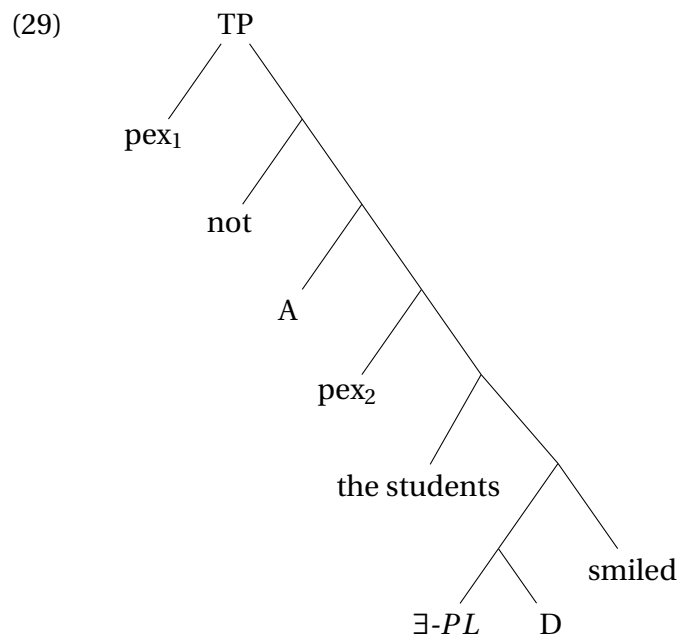
$$(28) \quad A(\llbracket (27\text{-a}) \rrbracket) = \neg\exists x \leq \llbracket \text{the students} \rrbracket : \llbracket \text{smiled} \rrbracket(x)$$

As for non-maximality, at this point we predict that there is no non-maximality under negation. Since the asserted component in (25-a) and (27-c) entails homogeneity, weakening the homogeneity presupposition in (25-b) by pruning does not affect the overall truth conditions after accommodation. In the next section, we argue that non-maximality is in fact possible under negation, but only if the homogeneity presupposition is accommodated under negation. This will account for the asymmetry in non-maximality between positive and negative environments: under negation, non-maximality requires local accommodation, which is known to be

a costly mechanism.

## 5 Non-maximality in negative sentences

While non-maximality readings are more difficult to obtain in downward-entailing environments, they are still available. This has been confirmed by experimental evidence, showing that non-maximal readings under a universal quantifier (*every*) are easier to obtain than under a negated existential quantifier (*no*) (Augurzky *et al.*, 2023). We therefore want to predict that non-maximal readings are, though (i) harder, (ii) not impossible to obtain in downward-entailing environments. In the following, we show that both (i) and (ii) can be explained within a *pex*-based account. In a nutshell, we will show that non-maximality under negation can be obtained if homogeneity is locally accommodated under negation, and then a second *pex* applies globally, as in (29). It has been argued that local accommodation is more difficult to obtain than global accommodation (Heim, 1983; Chemla and Bott, 2013). The asymmetry in non-maximality between positive and negative sentences can therefore be attributed under this account to the fact that in the latter non-maximality requires local accommodation.<sup>8</sup>



As argued in the previous section, *pex* is obligatory at every scope site.<sup>9</sup>

<sup>8</sup>Note that it remains an open question why local accommodation in fact seems to be easily available with homogeneity in other environments, where we do not see the expected projection behavior. For example, in order to account for the fact that homogeneity does not seem to project from the antecedents of conditionals, we have to posit that homogeneity is easily locally accommodated there. On the other hand, local accommodation seems to be more difficult in other environments, like the scope of non-monotonic quantifiers, and if this section is correct, under negation. This remains an open issue at this point and relates to the larger issue of how homogeneity projection differs in some environments from how presuppositions generally project (see section 6.1).

<sup>9</sup>As a reviewer points out, in (29), since we are assuming that **pex** should appear at every scope site, strictly speaking a **pex** should also appear between ‘not’ and the accommodation operator. However, given reasonable assumptions about how innocent inclusion works with *pex*, the contribution of this **pex** would be trivial here: A[**pex**[

This will ensure that the LF in (29) is the only available one when there is local accommodation below negation. The prejacent of the higher *pex* will have the *not-all* truth conditions brought about by accommodation. These will be strengthened to *not-some* via innocent inclusion of subdomain alternatives.

To see this step by step, consider first truth conditions for the prejacent of the higher *pex* in (30) (which are the negation of (20)).

- (30)  $\llbracket \text{not}[A[\text{pex}[\text{the students } \exists\text{-PL D smiled}]]] \rrbracket =$
- a. **asserts:**  
 $\neg \forall x \leq_{AT} \llbracket \text{the students} \rrbracket : x \in D \rightarrow \llbracket \text{smiled} \rrbracket (x)$
  - b. **presupposes:**  $\emptyset$

Now we want to compute assertion and presupposition of the parse with the global *pex*. To do this, we first need to examine what the alternatives to (30) are. As shown by Romoli (2012), considering alternatives in which negation is deleted leads to problems. We thus assume that the only alternatives for the higher *pex* are again the subdomain alternatives, with the effect that it again gives rise to the homogeneity presupposition in (31-b).

- (31)  $\llbracket \text{pex}_1[\text{not A} [\text{pex}_2[\text{the students } \exists\text{-PL D smiled}]]] \rrbracket^w$
- a. **asserts:**  
 $\neg \forall x \leq_{AT} \llbracket \text{the students} \rrbracket^w : x \in D \rightarrow \llbracket \text{smiled} \rrbracket^w (x)$  *modulo pruning*
  - b. **presupposes:**  
 $\forall \alpha. \alpha \in R \wedge \alpha \in \{\lambda w'. \neg \forall x \leq_{AT} \llbracket \text{the students} \rrbracket^w : x \in D' \rightarrow \llbracket \text{smiled} \rrbracket^{w'} (x) : D' \subseteq D\} \rightarrow \alpha(w) = 0$   
 $\models$   
 $\neg \exists x \leq_{AT} \llbracket \text{the students} \rrbracket^w : x \in D \wedge \llbracket \text{smiled} \rrbracket^w (x) \vee$   
 $\forall x \leq_{AT} \llbracket \text{the students} \rrbracket^w : x \in D \rightarrow \llbracket \text{smiled} \rrbracket^w (x)$  *modulo pruning*
  - c.  $\text{Alt}_{pex_1} = \{\lambda w'. \neg \forall x \leq_{AT} \llbracket \text{the students} \rrbracket^w : x \in D' \rightarrow \llbracket \text{smiled} \rrbracket^{w'} (x) | D' \subseteq D\}$

In case there is no pruning, the truth conditions after accommodation will be identical to (28). However, crucially, since the asserted component does not entail homogeneity, pruning is now able to weaken the meaning and therefore give rise to non-maximality. For simplicity, we consider here pruning only in the alternatives for the higher *pex* and show that we can get the

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the students smiled]]] simply asserts  $\forall x \leq_{AT} \llbracket \text{the students} \rrbracket : x \in D \rightarrow \llbracket \text{smiled} \rrbracket (x)$ . Considering sub-domain alternatives for this assertion, these are not innocently excludable, since they would contradict the assertion. On the other hand, the weaker subdomain alternatives are in fact innocently includable here, but the definition of *pex* given in Del Pinal et al. (2024) has to assume for independent reasons that weaker alternatives are not innocently includable. This is because otherwise in an example like (i-a), the alternatives *all* and *some* will be innocently includable and we would get a homogeneity presupposition with *all*, which is clearly undesirable. Note that the alternative definition we suggest in (17) avoids this stipulation, since the conditional presupposition is trivially met whenever the innocently includable alternatives are weaker than the prejacent. We therefore conclude that either way, this additional *pex* does not have an effect on our truth conditions.

- (i) a. All of the students smiled.
- b. Alternatives = { Some of the students danced, All of the students danced }

desired non-maximal readings.<sup>10</sup>

Consider the toy example in (32) with 3 students, and let's assume that only  $c$  smiled. We want to predict a non-maximal reading where *the students didn't smile* is true even though  $c$  did smile.

- (32) a.  $D = \{a, b, c\}$   
 b.  $\llbracket \text{student} \rrbracket = \{a, b, c\}$   
 c.  $\llbracket \text{smiled} \rrbracket = \{c\}$

Suppose we prune the alternative  $D' = \{c\}$ . Then, assuming a parse of “The students didn't smile” with accommodation, the homogeneity presupposition will be as in (33). Coupled with the *not-all* assertion, the truth conditions after accommodation are as in (34): (34) is true as long as everyone in the set  $\{a, b\}$  didn't smile. This is exactly the non-maximal reading we want to predict.

- (33)  $[\forall \alpha. \alpha \in R \wedge \alpha \in \{\lambda w'. \neg \forall x \leq_{AT} \llbracket \text{the students} \rrbracket^w : x \in D' \rightarrow \llbracket \text{smiled} \rrbracket^{w'}(x) : D' \neq \{c\} \wedge D' \subseteq D\} \rightarrow \alpha(w) = 1 \vee$   
 $\forall \alpha. \alpha \in R \wedge \alpha \in \{\lambda w'. \neg \forall x \leq_{AT} \llbracket \text{the students} \rrbracket^w : x \in D' \rightarrow \llbracket \text{smiled} \rrbracket^{w'}(x) : D' \neq \{c\} \wedge D' \subseteq D\} \rightarrow \alpha(w) = 0]$   
 (34)  $\neg \exists x \leq \llbracket \text{the students} \rrbracket : x \in D / \{c\} \wedge \llbracket \text{smiled} \rrbracket(x)$

As a result, given our assumption that *pex* applies at every scope site, the asymmetry in non-maximality is straightforwardly predicted: non-maximality is harder under negation since it requires local accommodation of the homogeneity presupposition below negation.<sup>11</sup>

## 5.1 Evidence from Križ and Chemla (2015)

An additional piece of evidence for asymmetries between positive and negative environments comes from Križ and Chemla (2015). In their experiment, participants were asked to either choose between *completely true* and *not completely true* or between *completely false* and *not completely false* for sentences that give rise to homogeneity such as ‘The triangles are(n't) green’.

<sup>10</sup>Notice that our new derivation of non-maximality doesn't hinge on *pex*, and something isomorphic is in fact achievable with *exh* as well, as follows:

- (i)  $\llbracket \text{The students didn't smile.} \rrbracket =$   
 $\text{exh}[\neg \text{exh}[\exists x. \text{stud}(x) \wedge \text{smiled}(x)]] =$   
 $\text{exh}[\neg \forall x \leq_{AT} \llbracket \text{the students} \rrbracket : x \in D \rightarrow \llbracket \text{smiled} \rrbracket(x)] =$   
 $\forall D'. D' \subseteq D \rightarrow [\neg \forall x \leq_{AT} \llbracket \text{the students} \rrbracket : x \in D' \rightarrow \llbracket \text{smiled} \rrbracket(x)] =$   
 $\neg \exists x \leq_{AT} \llbracket \text{the students} \rrbracket : x \in D \wedge \llbracket \text{smiled} \rrbracket(x)$  *modulo pruning*

Of course, this is not the end of the story: what is missing on an *exh*-based account is (i) why *exh* would be harder to embed under negation, i.e. what makes the LF above more difficult than one without *exh* under negation, and more importantly (ii) how *exh* captures the non-at-issueness of homogeneity.

<sup>11</sup>Note that the experimental evidence in Augurzky *et al.* (2023) considers definite plurals under negative quantifiers and not simply under negation. For reasons of space, we do not include a derivation of non-maximality under *no*, but just like with negation, local accommodation in the scope of *no* will be needed to derive non-maximal readings.

In conditions where not all of the objects had the target feature (e.g., being green for the sentence ‘The triangles are green’), if the sentence was a positive one, most participants judged it as *not completely false* and *not completely true*. On the other hand, for the negated counterpart, such as *The triangles aren’t green*, more than twice as many of the responses (around 50 percent) were *completely false*. This asymmetry between positive and negative environments also demands an explanation on our account.

If the *neither completely false nor completely true* responses are responses to presupposition failure (similarly to how the authors attribute them to truth value gaps), then on a *pex*-based account we would expect to have no asymmetry between positive and negative environments. In particular, as we saw above, we predict the same homogeneity presupposition in both cases.

However, Bar-Lev (2021) proposes an alternative explanation for the *neither completely true nor completely false* responses, which allows us to predict the asymmetry.<sup>12</sup>

(35) (from Bar-Lev, 2021)

Hypothesis: Gappiness is a side-effect of Non-maximality; ‘Neither true nor false’ is in fact ‘could be true and could be false’: could be true if the context justified a non-maximal reading, and could be false if it didn’t.

In a nutshell, whenever there is a pruning option that allows the sentence to be true, but it is not clear whether this option is licensed by the QUD, participants will judge the sentence as *not completely true* and also *not completely false*. Given that pruning is always available as an option to weaken the meaning in positive environments, we expect to then get overwhelming *neither true nor false* responses there. On the other hand, as we saw above, pruning can only weaken the meaning under negation if there is local accommodation below negation, which is costly.

Our contribution to the hypothesis in (35), then, is that the dispreferred local accommodation-based derivation of non-maximality in negative sentences also explains the reduced rate of *neither true nor false* judgments in negative sentences. In particular, we expect that in the gap scenarios, participants who do not locally accommodate will judge the sentence as *completely false*, thus explaining the higher rates of *completely false* under negation.<sup>13</sup>

## 5.2 Comparison with Paillé (2022)

At this point, two remarks are in order concerning the difference between Paillé’s original *pex*-based proposal and the one we just sketched. Paillé derives non-maximality from a trivalent denotation like (36) by following an idea from Križ (2015) where under certain QUDs, a sentence can be felicitously uttered even if its truth-value is undefined. Križ describes this notion as the proposition being true enough given the QUD. First, notice that the conditions of undefinedness of the sentences in (36) are exactly the same. This predicts that non-maximality should be completely symmetrical: if ‘Mary read the books’ can be true enough in a context where she read most but not all of the books, then ‘Mary didn’t read the books’ should be true

<sup>12</sup>Incidentally, attributing *neither completely true nor completely false* responses to presupposition failure would not align with Bassi *et al.*’s (2021) assumption that *pex* can be easily globally accommodated.

<sup>13</sup>We also thank an anonymous reviewer for encouraging us to discuss Križ and Chemla’s (2015) results and for suggesting an explanation of their results along the lines just described.

enough in a context where she read few but not none. This is precisely what seems not to be the case, if the evidence we have reviewed in section 2 is taken at face value.

$$(36) \quad \begin{array}{l} \text{a. } \text{pex}(\text{Mary read the books}) = \begin{cases} 1 & \text{if Mary read all of the books} \\ 0 & \text{if Mary read none of the books} \\ \# & \text{if Mary read only some of the books} \end{cases} \\ \text{b. } \text{pex}(\text{Mary did not read the books}) = \begin{cases} 1 & \text{if Mary read none of the books} \\ 0 & \text{if Mary read all of the books} \\ \# & \text{if Mary read some but not all of the books} \end{cases} \end{array}$$

Second, on the account proposed here, there is a principled reason why homogeneity gives rise to non-maximality while we don't see analogous weakening effects with standard presuppositions. This is because pruning is specific to implicatures and there is no analogous process for lexically triggered presuppositions. It is not clear how this asymmetry can be explained on Paillé's account of non-maximality: assuming that *pex* triggers a presupposition, we should see analogous 'true enough' effects with standard presuppositions. Of course, we could simply stipulate that only some truth-value gaps give rise to non-maximality, but our account avoids such a stipulation.

## 6 Further issues

### 6.1 Homogeneity projection

Turning to how homogeneity projects from embedded environments, Križ (2015) argues that homogeneity projects following a logic that only slightly differs from the Strong Kleene logic.<sup>14</sup> Independently, Bassi et al. (2021) show that assuming a Strong Kleene theory for presupposition projection, successfully derives certain data where implicature triggers are embedded (e.g. *some* under *some*). Therefore, if we assume a Strong Kleene theory of presupposition projection and that **pex** always applies at every scope cite (i.e. it applies at the lowest scope site), we predict roughly the same projection patterns for homogeneity as Križ (2015).

To illustrate this with an example, consider projection from the scope of non-monotonic quantifiers. In this environment, homogeneity projects as shown in (37), which is what is predicted by Strong Kleene after accommodating the presupposition globally. Turning to the **pex** theory, the LF in (38-a) is predicted. The embedded **pex** gives rise to a homogeneity presupposition which projects from the scope of *exactly two*. Following the Strong Kleene recipe, the truth conditions illustrated informally in (38-b) are predicted. After collapsing presupposition and assertion the correct truth conditions are predicted for (37).<sup>15</sup>

<sup>14</sup>For the purposes of this paper, we will ignore the differences between Križ's projection mechanism and Strong Kleene, but see Križ (2015) for a discussion of why the deviation from Strong Kleene is needed.

<sup>15</sup>This issue parallels a problem with FC in non-monotonic contexts; see Bar-Lev (2018, section 1.5) for non-trivial directions to resolve FC in such contexts, which would likewise apply to homogeneity in non-monotonic contexts as in (35).

- (37) Exactly two students read the books.  
 $\approx$  Exactly two students read all of the books and none of the other students read any books.
- (38) a. LF:  $[[\text{Exactly two students}] [1 [\text{pex}[t_1 \text{ read the books}]]]$   
 b.  $[[ (38\text{-a}) ]]$  = **presupposes**: (Exactly two students read all of the books  $\wedge$  No other student read any book)  $\vee$  (More than two students read all of the books)  $\vee$  (Less than two students read any books)  
**Asserts**: Exactly two students read at least 1 book.  
 c.  $A([ (38\text{-a}) ])=1$  iff Exactly two students read all of the books  $\wedge$  no other student read any book.

The same logic can be extended to definite plurals in other environments, making the same predictions as Križ's theory. Note that it is not clear how Bar-Lev's exhaustification theory is able to account for the truth conditions of (37).<sup>16</sup> If *exh* is inserted under the quantifier, as in (39), the inference that no other students read the books is not predicted. On the other hand, if *exh* is inserted above the quantifier, as in (40), the subdomain alternatives are innocently excludable and an unattested inference is predicted. For example, in the case where there are only two books *a* and *b*, the subdomain alternatives in (40-a) are innocently excludable and the truth conditions in (40) would be predicted. This is clearly an incorrect prediction, as (40) is predicted to be false if exactly two students read *a*, the same two students read *b* and no other students read any books.

- (39) Exactly two students  $1 \text{ exh}(t_1 \text{ read the books})= 1$  iff Exactly two students read all of the books.
- (40) *exh* (Exactly two students read the books).  
 a. Innocently Excludable alternatives (assuming there are only two books *a* and *b*):  
 Exactly two students read book *a*.  
 Exactly two students read book *b*.  
 b. Predicted truth conditions: Exactly two students read at least one book and it is not the case that exactly two students read book *a* and it is not the case that exactly two students read book *b*.

Since neither parse with assertive *exh* predicts the the correct truth conditions, we take this as another piece of evidence that favors adopting our **pex**-based theory for homogeneity rather than an assertive *exh*-based one.

Note that our account does not address the apparent differences in projection between homogeneity and standard presuppositions, for example in the antecedents of conditionals, as pointed out by Spector (2013) and Križ (2015). Fox (2018) tries to account for these differences within a presuppositional account of homogeneity by arguing that the homogeneity presupposition gives rise to a proviso problem in certain environments, leading to obligatory local accommodation. More generally, our account of homogeneity in terms of *pex* raises the question of whether the particular projection pattern of homogeneity in fact matches with that of scalar implicatures more generally, especially since homogeneity and *pex* have both been inde-

<sup>16</sup>Thanks to an anonymous reviewer for pointing this out.

pendently argued to project with a Strong Kleene Logic (Križ, 2015; Bassi et al., 2021). Ongoing research is exploring the differences in projection between *pex* and other presuppositions more generally in terms of redundancy (e.g. Wehbe, 2024), but we leave this issue aside for the purposes of this paper.

## 6.2 Free Choice

Our account relies on Del Pinal et al.’s (2024) definition of presuppositional exhaustification with innocent inclusion, which they use to account for free choice. In particular, Del Pinal et al. provide arguments that free choice behaves like a presupposition. If this is true, then we would expect free choice inferences to pattern with homogeneity with respect to the PAI diagnostics that we presented in section 3. In what follows, we show that surprisingly, free choice differs from homogeneity with respect to this test and discuss how we might make sense of these differences.

The predicted presupposition and assertion for a sentence with free choice are given in (41). Note that the presupposition in (41-b) is parallel to the homogeneity presupposition: (41) presupposes that either Mary can eat both cake and candy or she can eat neither.

- (41) *pex* (Mary can eat Cake or Candy)
- a. Assertion:  
 $\diamond$  (Cake or Candy)
  - b. Presupposition:  
 Either  $[\diamond$  (Cake) and  $\diamond$  (Candy)] or  $[\neg\diamond$  (Cake) and  $\neg\diamond$  (Candy)]

We can now set up parallel PAI examples to the ones with homogeneity above. Consider the example in (42). Here, the presupposition of the first sentence (that Mary can eat cake), coupled with the free choice presupposition that Mary can either eat both cake and candy or neither, makes the assertion in the second sentence trivial. This is because after accommodating the free choice presupposition of the second sentence, we end up in a common ground which entails that Mary can eat both cake and candy.

- (42) I knew that Mary can eat cake, but today I learned something interesting. She can eat cake or candy.

This contrasts starkly with the behavior of definite plurals, where analogous sentences are clearly infelicitous, as we have seen in examples like (10). Given the diverging behavior, we see three relevant theoretical possibilities: (i) Homogeneity is due to *pex* with innocent inclusion, but free choice isn’t; (ii) Free choice is due to *pex* with innocent inclusion, but homogeneity isn’t; (iii) Both are due to *pex* with innocent inclusion and some other mechanism explains the differences.

Given that the PAI data with homogeneity is robust and provides evidence for a presuppositional account, and given the asymmetries in non-maximality that we rely on, we believe that our evidence for a *pex*-based account of homogeneity still stands, regardless of how we explain the free choice data. This excludes option (ii).

This leaves us with options (i) and (iii). In other words, given that we take PAI effects as a test for whether there is innocent inclusion at the presuppositional level, we are committed to

free choice not being an ‘inclusion’ inference, unless additional elements explain the difference in PAI effects between homogeneity and free choice.<sup>17</sup> While we don’t take a stance on this, let us observe that there is one difference between the two phenomena that might help shed some light for future research on the source of the difference in PAI. The predicted homogeneity presupposition is not cancellable under negation, as shown in (43-b), while the free choice counterpart is, as shown in (43-a). In our *pex*-based account, the infelicity of (43-b) is predicted, since even if homogeneity is locally accommodated below negation, there is an obligatory higher *pex* that gives rise to the homogeneity inference again. It is not clear why free choice should differ in this respect. In both this case and the PAI tests, homogeneity patterns as expected given our *pex*-based account, but additional mechanisms seem to be needed to explain the divergence of free choice.

- (43) a. It is not the case that Mary can read {any book, War and Peace or Anna Karenina}.  
She can only read War and Peace.  
b. #It is not the case that Mary read the books. She only read War and Peace.

## 7 Conclusion

We have argued that an account based on presuppositional exhaustification provides a promising avenue for accounting for two empirical observations which, taken together, challenge all existing accounts of homogeneity and non-maximality: (i) homogeneity is sensitive to constraints on presupposition accommodation and (ii) non-maximal readings are more difficult to obtain in downward-entailing environments. Several details of the *pex*-based account remain to be worked out.

First, we only looked at cases where the definite plural was unembedded or embedded under negation. The predictions for how non-maximality arises when definite plurals are embedded in the scope of different quantifiers should be investigated in detail.

Additionally, looking beyond definite plurals, we observe that other phenomena which have been argued to give rise to homogeneity are also sensitive to PAI in the same way as definite plurals. Consider the examples below with homogeneity in other domains. If our analysis in terms of *pex* can be extended to these cases of homogeneity in other domains, we predict the infelicity of the second sentences in (44): in all of these cases, accommodating homogeneity makes the asserted component trivial. We can therefore use this as a diagnostic of whether an observed truth-value gap is in fact due to homogeneity.

- (44) a. **Habituals** (Agha, 2021):  
I already knew that Mary sometimes goes to the gym on Tuesdays, but today I learned something interesting. # Mary goes to the gym on Tuesdays.  
b. **Generics** (Löbner, 2000):  
I knew that lions sometimes hunt, but today I learned something interesting. # Lions hunt.

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<sup>17</sup>See also the conclusion, where we make a similar point about Neg-raising, which has been linked to homogeneity by some (Jeretic, 2022) .

- c. **Episodic universal bare plurals** (Condoravdi, 1996; Chierchia, 2022):  
I already knew that some birds are migrating right now, but today I learned something interesting. # Birds are migrating right now.
- d. **Predicate homogeneity** (Križ, 2015; Paillé, 2022):  
I already knew that the wall was partly red, but today I learned something interesting. # The wall is red.
- e. **Conditionals** (von Stechow, 1997):  
I already knew that sometimes, if you hit the brakes, the car stops, but today I learned something interesting. # If you hit the brakes, the car stops.

It remains an open question whether there can be a unified analysis of all these homogeneity phenomena in terms of *pex*, but at first sight it seems promising that all of these cases show the same pattern with respect to PAI.

To conclude, let us note that some cases that have been analyzed as homogeneity fail the PAI diagnostic. This makes it even clearer that PAI is useful as a diagnostic for the truth-value gap that homogeneity gives rise to.<sup>18</sup>

Going beyond the point we already made about free choice, take the example of neg-raising, which has been argued to be a homogeneous phenomenon (Jertič, 2021 and Staniszewski, 2022). This phenomenon does not pattern with the other cases of homogeneity discussed in (44). For example, consider *should*, which seems to exhibit a truth-value gap, as shown in (45). If we want to analyze this as a homogeneity effect, we expect within the **pex** account that (a) and (b) have the homogeneity presupposition given in (46). We can now test whether this is in fact a presupposition of the sentence with PAI. Consider the example in (47): here even though the first sentence, coupled with homogeneity, makes the second sentence trivial, the second sentence is felicitous. This raises questions about the analysis of neg-raising in terms of homogeneity.<sup>19,20</sup>

- (45) a.  $\llbracket \text{Mary should read this book} \rrbracket$  is True iff for **all** worlds compatible with Mary's obligations, she reads this book.
  - b.  $\llbracket \text{Mary shouldn't read this book} \rrbracket$  is True iff for **no** worlds compatible with Mary's obligations does she read this book.
  - c. **Neither is True** iff in only some worlds compatible with Mary's obligations, she reads this book.
- (46) **Homogeneity presupposition (should):** Either for all worlds *w* compatible with Mary's obligations, she reads the book in *w* or for no worlds *w* compatible with Mary's obligations, she reads the book in *w*.

<sup>18</sup>We thank a reviewer for pointing this out.

<sup>19</sup>Notice that it is crucial to look at negative cases, too, since positive cases would produce fine sentences if the neg-raising verb were underlyingly universal. It is therefore crucial that negative cases not only produce truth conditions that are clearly those of a negated existential, but also yield felicitous sentences (cf. (47-b)).

<sup>20</sup>Notice that examples parallel to (43) pattern like free choice and unlike homogeneity (Bar-Lev, p.c.).

- (i) It is not the case that Mary should read this book. She is allowed to, but it's also fine if she doesn't read it.

Bar-Lev, p.c., points out that both in (43) and in (i), stress seems to be required on respectively the free choice item and *should*, which might support the alternative, focus-based theory of some of our data.

- (47)
- a. I knew that Mary is allowed to read this book, but today I learned something interesting. She should read this book.
  - b. I knew that Mary is allowed not to read this book, but today I learned something interesting. She shouldn't read this book.

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